

Short communication

Exceptional high number of IPD cases in winter season 2024–2025 in Belgium in concomitance with rise in vaccine serotypes

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ABSTRACT

Following lifting of containment measures post-COVID, the number of invasive pneumococcal disease (IPD) infections continued to increase, reaching a record number last winter season 2024–2025 in Belgium. This exceptional increase is largely due to high numbers in (older) adults, and mainly an increase of vaccine serotype IPD. Over 95 % of IPD cases last winter could theoretically be prevented by vaccination. Cases of serotypes 12F, 14 and 4 were three or more times higher compared to pre-COVID winter season 2019–2020.

1. Brief description of the event and the aim of the study

The number of invasive pneumococcal disease (IPD) infections has evolved over time, impacted by both the implementation and lifting of non-pharmaceutical interventions in the context of COVID-19. Post-COVID, IPD cases continued to increase, reaching a record number of 1681 IPD isolates last winter season (October to March) 2024–2025 in Belgium. Pre-COVID, the national reference centre for IPD received around 1500 IPD isolates over the course of an entire year, or 992 IPD isolates for winter season 2018–2019, contributed by a stable number of laboratories located across the entire country. The exceptional increase in IPD cases during last winter season 2024–2025 was investigated in detail to pinpoint the affected age groups and to map the distribution of IPD serotypes.

2. Record number of IPD infections during winter season 2024–2025

Due to the restrictive measures in the context of COVID-19, an important reduction in the number of invasive pneumococcal disease (IPD) infections was reported worldwide [1,2], pronounced in Belgium as a decrease of 69.2 % in overall IPD cases for winter season (October–March) 2020–2021 compared to pre-COVID winter season 2018–2019 (decrease of 992 to 306 IPD infections; Fig. 1A). The largest reduction was observed amongst the group of older adults (≥ 65 years

old), with a decrease of nearly 79 % in the number of infections between winter seasons 2018–2019 and 2020–2021 (536 to 114 IPD infections). Following lifting of non-pharmaceutical interventions post-COVID, preceded by a period with limited exposure to respiratory pathogens and hence and anticipated immunity decrease, the number of IPD infections started to increase again from winter season 2022–2023 on, more specifically an increase of 73.5 % compared to winter season 2020–2021 (306 to 1153 IPD infections), exceeding pre-COVID numbers by 14.0 %. In accordance to the largest reduction during the COVID-19 pandemic, this increase of infections post-COVID was most pronounced in the older population (≥ 65 years old), resulting in more than 5 times the number of IPD infections in winter season 2022–2023 compared to 2020–2021. The next season 2023–2024, the number of IPD isolates received at the national reference centre (NRC) for invasive pneumococci in Belgium, remained overall stable (+2.1 %; 1178 IPD cases), while specifically for children (<18 years old) a further increase of nearly 18 % was noted. Remarkably, for last winter season 2024–2025, an overall IPD increase of 29.9 % was noted compared to 2023–2024, exceeding case numbers from all prior seasons, reflecting an increase of 41.0 % compared to pre-COVID. This exceptional increase is largely due to high numbers in adults (18–64 years; +59.1 %) and older adults (≥ 65 years; +43.7 %). For a period of six months during winter season 2024–2025 (October to March), an absolute number of 1681 IPD cases was reported, exceeding numbers of pre-COVID years 2017–2019 (January to December) as the yearly number of IPD isolates (12 months

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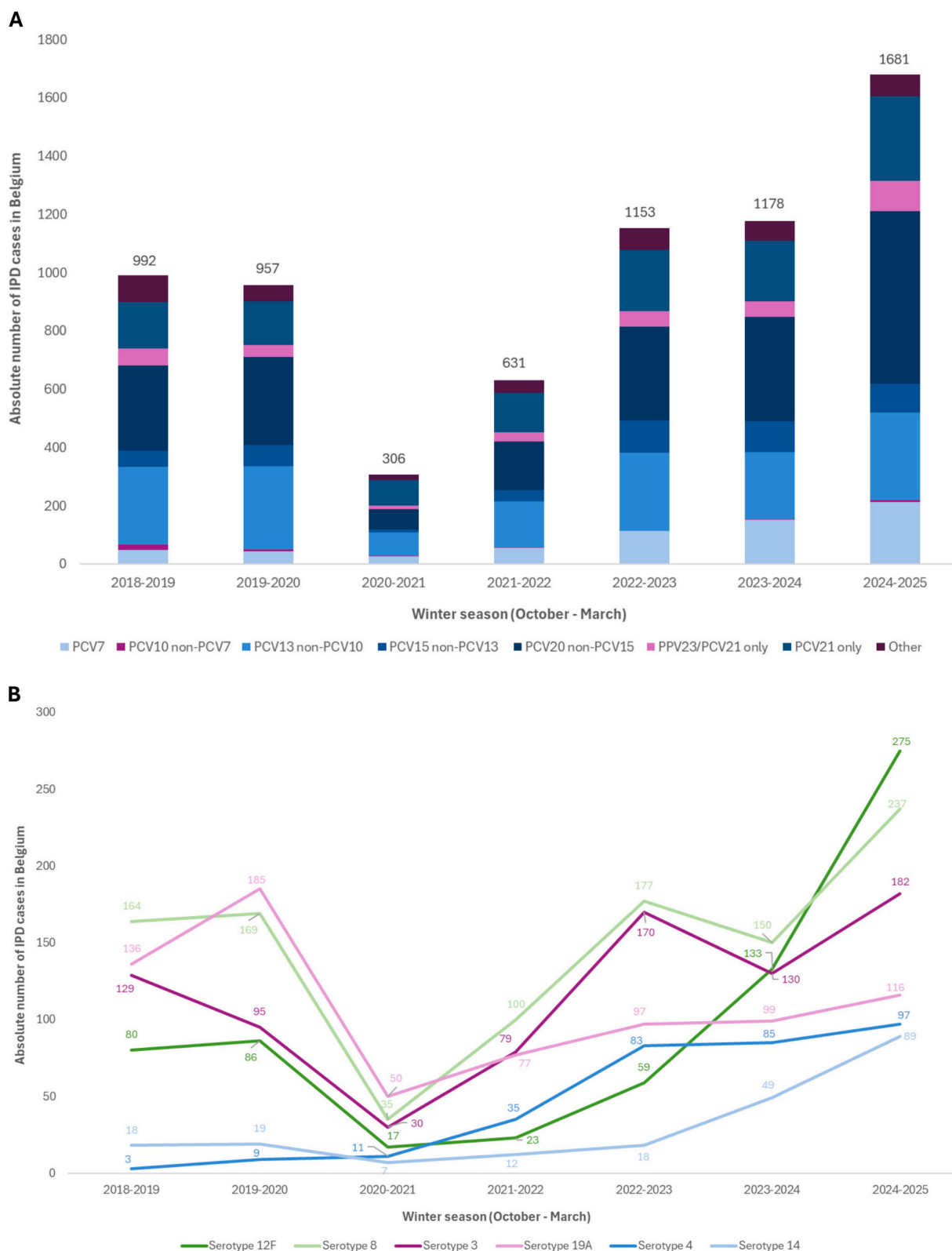


Fig. 1. A) Absolute number of IPD cases in Belgium for winter seasons (October to March) 2018-2019 to 2024-2025. Vaccine serotypes (as included within PCV7, PCV10, PCV13, PCV15, PCV20, PPV23 and PCV21*) and other serotypes are stacked. B) The six IPD serotypes causing > 5 % of total IPD infections in 2024-2025 (all vaccine serotypes) are plotted over the course of time (2018-2019 to 2024-2025), with the absolute case numbers per serotype visualised per winter season. * PCV21 has not yet been implemented in the national vaccination program of Belgium.

period) received at the NRC in Belgium ranged around 1500–1600.

3. Rise in vaccine serotype IPD

Interestingly, the increase in IPD is mainly due to an increase of vaccine serotype IPD (e.g. IPD caused by serotypes included in one of pneumococcal vaccines approved in 2025, however not all used in Belgium), accounting for 95.5 % of infections in 2024–2025 (1605/1681; Fig. 1A). While theoretical coverage of vaccine serotypes was already >90 % in winter season 2018–2019, the absolute number of IPD infections caused by PCV7 serotypes has increased by four-fold between the respective winter seasons 2018–2019 and 2024–2025. The top 6 serotypes causing IPD in the most recent winter season 2024–2025 are all vaccine serotypes: PCV20 non-PCV15 serotypes 12F and 8, PCV13 non-PCV10 serotypes 3 and 19 A, and PCV7 serotypes 4 and 14, respectively representing 16.4 %, 14.1 %, 10.8 %, 6.9 %, 5.8 % and 5.3 % of overall IPD cases in 2024–2025. All these serotypes were heavily reduced between 2018–2019 and 2020–2021 (ranging between –61.1 % and –78.8 %), except for serotype 4 that started to increase (Fig. 1B). The largest increase in serotype 4 (nearly 8-fold) was seen between winter seasons 2020–2021 and 2022–2023, with most cases diagnosed in adults (18–64 years) [3]. During last two winter seasons 2023–2024 and 2024–2025, drastic increases in case numbers were observed for serotypes 12F and 14 across all age groups, while most pronounced in (older) adults (≥65 years old). Cases of serotypes 12F, 14 and 4 were respectively three, nearly five and more than ten times higher in winter season 2024–2025 compared to pre-COVID winter season 2019–2020. Except for PCV13 non-PCV10 serotype 19 A for which lower absolute case numbers are observed post-COVID compared to pre-COVID (136 IPD infections for 2018–2019 versus 116 for 2024–2025), for all other serotypes part of the top 6 serotypes causing IPD in 2024–2025, the number of IPD infections has increased remarkably.

4. Importance of stable and longstanding IPD surveillance

Vaccine uptake (>90 %) and vaccine product used (PCV13 – Prevenar 13®) in children remained unchanged in Belgium since 2019 [4,5]. Despite a change in recommendation from PCV13 + PPV23 (Prevenar 13® + Pneumovax 23®) to PCV20 (Prevenar 20®) for adults in 2022 and the (partial) reimbursement of PCV20 since 2024, adult vaccine uptake remained stable and low (<25 %) in adult risk groups [6]. Therefore, immunisation policy is not considered as one of main drivers of this observed change in epidemiology. While IPD surveillance as conducted by the Belgian NRC is a passive laboratory-based surveillance, it remained stable over time with approximately 90 laboratories spread across Belgium participating every year by sending isolates from invasive sources (defined as isolation of *S. pneumoniae* from normal sterile body compartments) to the NRC, estimating the coverage of the surveillance to be >90 % [7]. Moreover, also no changes in blood culture practices in Belgium were observed over the studied period (2018–2025) (data not shown). All IPD isolates at the NRC undergo capsular typing by Quellung followed by antimicrobial susceptibility testing.

COVID-19 and its containment measures most likely played a crucial role, as adults may have been less in contact with pneumococci during that period, resulting in decreased natural immunity in the population. However, the impact thereof would have expected to be already observed in the first winter season (2022–2023) following the end of the containment measures in the context of the pandemic in January 2022. While overall IPD was drastically affected by these containment measures, a sustained and high infant pneumococcal carriage rate was observed during periods of stringent restrictive measures [8]. Other contributing factors may be the emergence of new pneumococcal clones characterised by increased and concomitant increase of viral infections. However, surveillance of (severe) respiratory infections in Belgium did not detect an exceptional winter season peak in 2024–2025 compared to

pre-COVID years [9].

5. Conclusion

Belgian data warrant for close monitoring of the number of IPD cases as evolutions pre-COVID and post-COVID seem to differ, stressing the importance of long-term national surveillance programs [10]. Our results underscore the need for implementation of higher valency vaccines and awareness of the increase of some PCV13 serotypes despite the inclusion of these serotypes in immunisation programmes since many years.

CRedit authorship contribution statement

Lize Cuypers: Writing – original draft, Visualization, Formal analysis, Data curation, Conceptualization. **Cato Dambre:** Writing – review & editing, Data curation. **Stefanie Desmet:** Writing – original draft, Supervision, Conceptualization.

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Lize Cuypers reports a relationship with Merck & Co Inc that includes: travel reimbursement. Stefanie Desmet reports a relationship with Merck & Co Inc that includes: board membership and speaking and lecture fees. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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